

Windows VM Migration

How hyper2kvm Safely Migrates Windows from VMware, Hyper-V, Azure & AWS to KVM

The only open-source tool that handles the complete Windows migration pipeline: VirtIO driver injection, registry fixes, BSOD prevention, BCD repair, VMware/Hyper-V tools removal, and automated boot verification. Supports Windows Server 2016-2025 and Windows 10/11 across BIOS and UEFI.

Zero BSOD | Auto VirtIO | SQL Server & AD Safe | Boot Verified | Apache 2.0

The Windows BSOD Problem

Why Windows VMs crash after migration — and how hyper2kvm prevents every failure mode.

Without proper driver injection, every Windows VM will BSOD on first boot

Windows loads storage drivers at boot from the registry. VMware uses LSI Logic/PVSCSI, Hyper-V uses storvsc, AWS uses Xen/NVMe. KVM uses VirtIO. If VirtIO drivers aren't pre-installed in the offline disk image, Windows cannot access the boot volume and crashes with **INACCESSIBLE_BOOT_DEVICE (0x7B)**. This is the #1 Windows migration failure — hyper2kvm eliminates it completely.

Every Failure Mode — Automatically Fixed

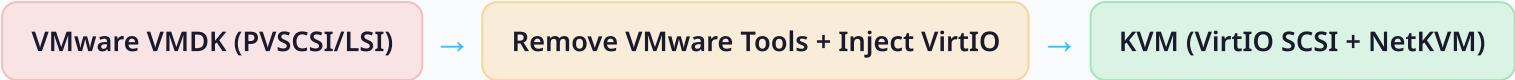
Failure	BSOD / Error	Root Cause	hyper2kvm Automatic Fix
Boot disk invisible	0x7B INACCESSIBLE_BOOT_DEVICE	Missing VirtIO storage driver	Injects viostor/vioscsi from virtio-win ISO into offline disk
No network	No adapter found	VMware VMXNET3 or Hyper-V netvsc gone	Injects NetKVM (VirtIO network driver) + configures registry
Memory pressure	High RAM usage	No balloon driver for dynamic memory	Injects Balloon driver for KVM memory management
Slow disk I/O	Performance degradation	Fallback to emulated IDE	Configures VirtIO SCSI controller with proper registry entries
VMware Tools crash	Service errors on boot	VMware agents can't find hypervisor	Removes VMware Tools services, drivers, and tray applet
Hyper-V IC conflict	Driver load errors	Hyper-V Integration Components present	Removes Hyper-V IC drivers and services from registry
AWS agent conflict	SSM/CloudWatch errors	AWS agents can't reach metadata	Disables AWS SSM, CloudWatch, EC2Config services
Boot config wrong	Boot failure	BCD references old controller path	Updates BCD entries for VirtIO disk controller
Wrong HAL	0x5C HAL_INITIALIZATION_FAILED	ACPI/APIC mismatch	Configures correct HAL for KVM machine type (i440fx/q35)
Serial console missing	No headless access	Serial not configured	Enables COM1 serial console for remote management

All fixes are applied offline — hyper2kvm mounts the disk image via qemu-nbd, modifies the Windows registry and filesystem directly, then unmounts. The VM never needs to boot in a broken state.

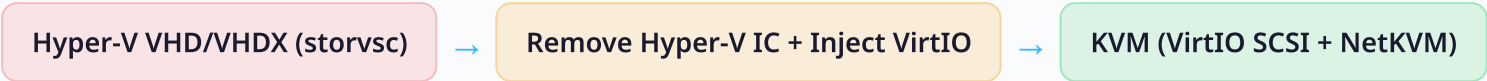
Every Source Platform

hyper2kvm migrates Windows VMs from VMware, Hyper-V, AWS, Azure, and more — each with source-specific fixes.

VMware vSphere → KVM



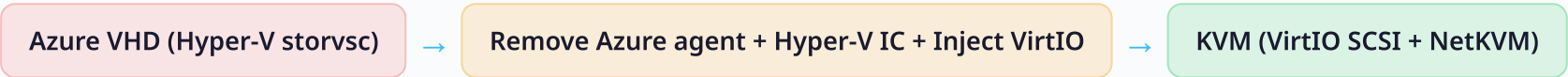
Hyper-V → KVM



AWS EC2 → KVM



Azure VM → KVM



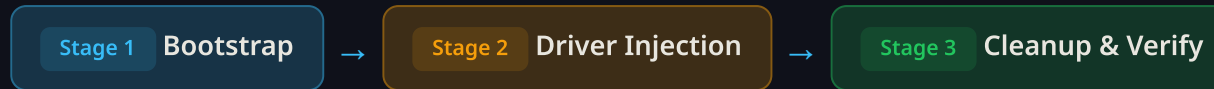
Source-Specific Fix Matrix

Fix Applied	VMware	Hyper-V	AWS EC2	Azure
VirtIO storage driver	✓	✓	✓	✓
VirtIO network driver	✓	✓	✓	✓
VirtIO balloon	✓	✓	✓	✓
Remove VMware Tools	✓	N/A	N/A	N/A
Remove Hyper-V IC	N/A	✓	N/A	✓
Remove cloud agents	N/A	N/A	✓ (SSM, EC2Config)	✓ (Azure agent)
BCD boot fix	✓	✓	✓	✓

Registry driver stack	✓	✓	✓	✓
Serial console enable	✓	✓	✓	✓

Multi-Stage Pipeline

hyper2kvm uses a 3-stage approach for safe, reliable Windows migration.



Stage 1 Bootstrap

- Mount Windows disk offline via qemu-nbd
- Detect Windows version (2016/2019/2022/2025/10/11)
- Detect boot mode (BIOS vs UEFI)
- Detect current storage controller (LSI/PVSCSI/storvsc)
- Locate Windows registry hives (SYSTEM, SOFTWARE)
- Identify installed hypervisor tools
- Map partition layout (EFI System, Recovery, C:)
- Select correct VirtIO driver version for OS

Stage 2 Driver Injection

- Extract VirtIO drivers from virtio-win ISO
- Copy viostor/vioscsi .sys + .inf to C:\Windows\drivers
- Copy NetKVM .sys + .inf for network
- Copy Balloon .sys + .inf for memory
- Load offline SYSTEM hive, add driver registry entries
- Set VirtIO storage driver to boot-start (Start=0)
- Update CriticalDeviceDatabase for VirtIO PCI IDs
- Configure UpperFilters/LowerFilters for storage stack

Stage 3 Cleanup & Verify

- Remove VMware/Hyper-V/cloud agent services
- Remove source hypervisor drivers from driver stack
- Update BCD for new boot disk controller
- Enable serial console (COM1) for headless access
- Generate libvirt XML with VirtIO + Q35 machine
- Generate KubeVirt manifest (optional)
- Automated boot test: start VM, verify no BSOD
- Webhook notification on success/failure

YAML Configuration — Full Windows Migration

```
# windows-migration.yaml — Migrate Windows Server 2022 from VMware to KVM
cmd: vsphere                                # Or: local, hyperv
vcenter: vcenter.corp.com                  # vSphere source (if cmd: vsphere)
vs_action: export_vm
vs_vm: win-sql-server-01
output_dir: ./converted
out_format: qcow2
windows: true                               # Enable Windows migration pipeline
guest_os: windows
virtio_win_iso: /usr/share/virtio-win/virtio-win.iso
win_stage: bootstrap                        # Stage 1 → 2 → 3 automatic
emit_domain_xml: true                       # Generate libvirt domain XML
deploy_k8s: false                           # Or true for KubeVirt deploy
vm_name: win-sql-server-01
```

```
memory: 8192      # 8 GB RAM
vcpus: 4
machine: q35      # Modern chipset with PCIe
uefi: true        # UEFI boot (or false for BIOS)
boot_test: true   # Verify VM boots without BSOD
```

Supported Windows Versions

Every modern Windows edition, BIOS and UEFI, tested and verified.

Operating System	Editions	Boot Mode	Status	Notes
Windows Server 2025	Standard, Datacenter	UEFI, BIOS	Tested	Latest VirtIO drivers, Q35 machine type recommended
Windows Server 2022	Standard, Datacenter, Core	UEFI, BIOS	Tested	Primary target for enterprise migration
Windows Server 2019	Standard, Datacenter	UEFI, BIOS	Tested	Most common in current VMware estates
Windows Server 2016	Standard, Datacenter	UEFI, BIOS	Tested	Extended support until 2027, migrate now
Windows 11	Pro, Enterprise	UEFI + TPM	Tested	TPM 2.0 emulated via swtpm on KVM
Windows 10	Pro, Enterprise, LTSC	UEFI, BIOS	Tested	VDI and workstation migrations
Windows Server 2012 R2	Standard, Datacenter	BIOS	Tested	Legacy — use older VirtIO driver set

Enterprise Windows Workloads — Tested & Safe

SQL Server

- SQL Server 2016, 2017, 2019, 2022 — all tested
- Database files migrate with disk — no reinstall
- VirtIO SCSI matches PVSCSI I/O performance
- TempDB, log files, data files all intact
- **Licensing win:** KVM vCPU pinning can reduce per-core license costs by 20-40%
- Always-On AG: migrate secondary replicas first

Active Directory Domain Controllers

- NTDS.dit database migrates with disk
- SYSVOL/NETLOGON shares preserved
- AD replication resumes after boot on KVM
- **Strategy:** Migrate secondary DCs first, verify replication, then primary
- DNS zones and records intact
- MAC address optional preservation for DHCP

Exchange Server

- Exchange 2016, 2019 — tested successfully
- Mailbox databases mount correctly post-migration

IIS / Web Applications

- IIS sites, app pools, bindings all preserved
- SSL certificates in Windows cert store intact

- Put in maintenance mode during cutover window
- DAG members: migrate passive copies first
- Certificate stores preserved in registry
- Transport queues drain before migration

- .NET applications run unchanged
- Simplest migration — standard VirtIO injection
- Verify URL bindings and DNS post-migration
- Load balancer health checks resume automatically

File Servers (NTFS/ReFS)

- NTFS permissions (ACLs) fully preserved
- DFS namespaces and replication intact
- SMB shares auto-start on boot
- Deduplication volumes migrate correctly
- Shadow copies preserved with disk
- FSRM quotas and file screens maintained

Remote Desktop Services (RDS)

- Session host, broker, gateway roles migrate
- User profiles and settings preserved
- RDP certificates in store intact
- Connection broker database migrates with SQL
- GPU passthrough possible on KVM for RemoteFX
- Licensing server: reactivate after migration

SQL Server Licensing Savings

Migrating Windows + SQL Server from VMware to KVM can dramatically reduce Microsoft licensing costs.

20-40%

SQL Server license savings
with KVM vCPU pinning

\$242K

SQL Enterprise per-server cost
that right-sizing can reduce

\$0

Hypervisor license cost on KVM
(vs \$6K-\$15K VMware per host)

2x

License savings stack:
no VMware + reduced SQL cores

How KVM Reduces SQL Server Licensing

The VMware Problem

- VMware vSphere reports all physical cores to Microsoft
- SQL Server per-core licensing counts cores on the host
- A 2-socket, 16-core host = 32 cores to license
- Even if SQL VM uses only 4 vCPUs, you may need to license all 32 cores
- SQL Server Standard: \$3,945/core = **\$126,240 per host**
- SQL Server Enterprise: \$15,123/core = **\$483,936 per host**

The KVM Advantage

- KVM with CPU pinning exposes only assigned cores
- SQL VM with 4 pinned vCPUs = license only 4 cores
- NUMA-aware placement for performance + compliance
- cgroups enforce CPU isolation — verifiable by auditors
- SQL Server Standard: \$3,945 x 4 = **\$15,780**
- SQL Server Enterprise: \$15,123 x 4 = **\$60,492**

Cost Comparison: 10 SQL Server VMs (3-Year TCO)

Cost Item	VMware vSphere	KVM (hyper2kvm)	Savings
Hypervisor licensing	\$150,000 (3 yr)	\$0	\$150,000
SQL Standard (10 VMs)	\$1,263,600	\$631,200 (pinned)	\$632,400

Windows Server	\$120,000	\$120,000	\$0
Migration cost	N/A	\$10,000 (one-time)	-\$10,000
Support / ops	\$90,000	\$75,000	\$15,000
3-Year Total	\$1,623,600	\$836,200	\$787,400

\$787K Saved — SQL Server Alone

Eliminate VMware licensing (\$150K) + reduce SQL per-core costs (\$632K) by migrating to KVM with hyper2kvm.
The migration tool is free. The savings are real.

Boot Verification & Safety

Every Windows migration is verified before production cutover. Zero risk.

Automated Boot Test

After conversion, hyper2kvm starts the Windows VM on KVM, monitors boot progress, and confirms it reaches the login screen without BSOD. **No human needs to watch.** If boot fails, you're notified immediately with the error code.

VNC/noVNC Console

Full graphical console available for visual verification. See exactly what Windows displays on first boot. Debug any issues in real-time. Connect from any browser via noVNC — no client software needed.

Post-Boot Health Checks

- Disk I/O test — VirtIO SCSI responding
- Network connectivity — NetKVM driver active
- Windows services running (SQL, AD, IIS...)
- Event log scan for critical errors
- RDP accessible on port 3389
- Serial console functional on COM1

Zero-Risk Rollback

- Source VMDK/VHD preserved until verification passes
- Original VM untouched on VMware/Hyper-V/cloud
- Conversion creates a copy — never modifies source
- If boot test fails, fix and retry — source is safe
- Webhook alerts team on success or failure
- Auditable pass/fail report per VM

Migration Safety Checklist — What hyper2kvm Validates

#	Check	What It Validates	Why It Matters
1	Pre-flight (doctor)	Disk space, qemu-nbd, virtio-win ISO, libvirt	Don't start if dependencies are missing
2	Source disk integrity	VMDK descriptor, extent files, partition table	Catch corrupt disks before wasting hours
3	Windows version detect	Registry: CurrentVersion, CurrentBuild	Select correct VirtIO driver set
4	Boot mode detect	EFI System Partition present?	UEFI vs BIOS affects BCD and XML config
5	Driver injection verify	viostor.sys exists + registry Start=0	Confirm driver will load at boot
6	Source tools removed	VMware/Hyper-V services disabled in registry	Prevent conflicts and error spam
7	Boot test	VM reaches login screen, no BSOD	The ultimate validation — it works

Compare with manual migration: Without hyper2kvm, operators discover boot failures only after starting the VM. For a remote SQL Server, that could mean hours of downtime debugging a BSOD at 2 AM. hyper2kvm catches every issue *before* the cutover window.

Windows Migration Tool Comparison

How hyper2kvm compares to other tools for Windows VM migration.

Capability	hyper2kvm	virt-v2v	Red Hat MTV	AWS MGN	Manual
VirtIO driver injection	✓ Auto	✓ Auto	✓	N/A	Manual
Registry driver fix	✓ Auto	✓	Basic	N/A	Manual
VMware Tools removal	✓ Auto	✓	Partial	N/A	Manual
Hyper-V IC removal	✓ Auto	✓	✗	N/A	Manual
AWS agent removal	✓ Auto	✗	✗	N/A	Manual
BCD boot repair	✓ Auto	✓	Basic	N/A	Complex
Boot verification	✓ Auto	✗	✗	✗	Manual
UEFI support	✓	✓	✓	N/A	Complex
TPM emulation	✓ swtpm	✗	✓	N/A	Manual
Deploy to KVM/libvirt	✓	✓	✗	✗	Manual
Deploy to KubeVirt	✓	✗	✓	✗	✗
Interactive TUI	✓ zkvm	✗	✗	✗	✗
Batch migration	✓	✗	✓	✓	✗
No libguestfs needed	✓ VMCraft	Required	Required	N/A	N/A
Source: VMware	✓	✓	✓	✓	Manual
Source: Hyper-V	✓	✓	Partial	✓	Manual
Source: AWS/Azure/GCP	✓	✗	✗	AWS only	Complex
Cost	\$0	\$0	\$50K+ (OCP)	Free/90d	\$0

hyper2kvm is the only tool that: (1) migrates Windows from VMware + Hyper-V + AWS + Azure + GCP, (2) deploys to both KVM and KubeVirt, (3) includes automated boot verification, (4) requires no libguestfs, and (5) costs \$0.

Real-World Scenarios

Enterprise Windows migration projects powered by hyper2kvm.

Scenario 1: VMware Exit — 50 Windows Servers

- **Source:** vSphere 7, 50 Windows Server 2019/2022 VMs
- **Workloads:** 10 SQL, 5 AD DCs, 15 IIS, 10 file servers, 10 app
- **Target:** KVM on RHEL 9 bare-metal hosts
- **Process:** zkvm TUI discovers VMs, batch export, auto VirtIO injection, boot verification
- **Timeline:** 3 weeks (10 VMs/week, off-hours cutover)
- **Savings:** \$300K/yr VMware + \$632K SQL licensing = **\$932K/yr**

Scenario 2: Azure Repatriation — 20 Windows VMs

- **Source:** Azure D-series VMs, Windows Server 2022
- **Workloads:** .NET web apps, SQL Server, file shares
- **Target:** On-prem KVM (data sovereignty requirement)
- **Process:** hypersdk exports managed disks, hyper2kvm removes Azure agent + Hyper-V IC, injects VirtIO, boot test
- **Timeline:** 2 weeks
- **Savings:** \$480K/yr in Azure compute + egress

Scenario 3: Hyper-V to KubeVirt — 30 Windows VMs

- **Source:** Hyper-V 2019 cluster, Windows 10 VDI + servers
- **Workloads:** VDI desktops, RDS session hosts, app servers
- **Target:** K3s + KubeVirt (modernization initiative)
- **Process:** VHD export, remove Hyper-V IC, inject VirtIO, generate KubeVirt manifests, kubectl apply
- **Timeline:** 2 weeks
- **Savings:** \$100K/yr Hyper-V + Windows Datacenter licensing

Scenario 4: Multi-Cloud Windows Consolidation

- **Source:** 10 on AWS, 8 on Azure, 12 on VMware — all Windows
- **Workloads:** Mixed enterprise apps, SQL, web, services
- **Target:** On-prem KVM cluster (single management plane)
- **Process:** hypersdk exports from all 3 sources, hyper2kvm applies source-specific fixes (AWS agents, Azure agents, VMware Tools), unified KVM deploy
- **Timeline:** 4 weeks
- **Savings:** \$800K/yr cloud + VMware combined

Windows Migration Made Safe and Simple

From any source. Full VirtIO injection. Boot verified. Zero BSOD.

SQL Server, Active Directory, Exchange, IIS — all tested, all safe.

hyper2kvm: the only tool that handles every Windows migration path.

